

Student Mentoring and Tracking System for Indian Schools: Ecosystem Analysis and Detailed SRS

Indian school ecosystem and constraints that will shape your product

Indian schools are structurally diverse: different management types (government, aided, private unaided), different operating models (day schools, residential schools), and different exam boards. Any mentoring/tracking product that assumes a single “one-size-fits-all” process will break quickly in the field. The safest design approach is to be **board-agnostic** while allowing configuration for local roles, reporting formats, and language needs.

A key system-level change driving “tracking” requirements is that National Education Policy 2020 ¹ proposes a **5+3+3+4** curricular and pedagogical structure covering ages 3–18, replacing the older 10+2 framing (even though many schools still use “Class 1–12” operationally). ² The same policy explicitly pushes assessment away from rote/summative patterns toward **regular, formative, competency-based** assessment. ³

Most importantly for your idea, NEP 2020 is explicit that the student progress card should be redesigned into a **holistic, 360-degree, multidimensional report** including **self-assessment, peer assessment, portfolios, project work**, and teacher assessment—and it even calls out that **AI-based software could be developed** to help track growth over school years. ³ This is basically a policy-backed justification for your “complete tracking” direction—provided you implement it with strong safeguards.

On the operational/data side, India already has a nationwide school data backbone: UDISE+ ⁴ (Department of School Education & Literacy, Ministry of Education ⁵) is positioned as a large education management information system and is intended to be online and move toward “real time” data collection. ⁶ It also exposes a practical reality you must design for: in remote areas, processes may involve offline capture with later uploading, and systems rely heavily on validation checks and block-level support workflows. ⁷

You should treat “UDISE code” as a first-class identifier at the school level: the UDISE reporting module states that the **UDISE code of the school is the username** (with credentials managed via block/district MIS support). ⁸ This strongly suggests your onboarding and imports should support “school-by-UDISE-code” mapping even if you don’t integrate directly with government systems.

Private-board compliance adds more constraints. For example, Central Board of Secondary Education ⁹ affiliation bye-laws require schools to maintain specific records and preserve exam records, and they reference annual e-returns on OASIS and U-DISE portals. ¹⁰ CBSE’s teacher expectation guidance also explicitly expects record-keeping in a teacher diary for teaching and pedagogical activities. ¹¹ Those

compliance norms matter because your product will be judged on whether it makes recordkeeping easier (and audit-safe) rather than creating an additional burden layer.

Counseling is no longer “optional” in many contexts. CBSE circulars (Jan 2026) amend affiliation expectations so secondary/senior secondary CBSE schools must appoint counseling/wellness and career counseling roles with a stated **1:500** ratio, with scope for alternatives and training expectations. ¹² CBSE has also formalized a **Hub & Spoke school counseling model** for strengthening school psychosocial support through collaboration and monthly reporting. ¹³ Separately, National Council of Educational Research and Training ¹⁴ guidance documents describe statewide and school-level approaches to implementing guidance and counseling programs, including structures and responsibilities. ¹⁵

Finally, the **privacy and child-safety bar is high**. The Digital Personal Data Protection Act defines a child as under **18** and treats the parent/guardian as included in “Data Principal” for a child. ¹⁶ It requires **verifiable parental consent** before processing a child’s personal data, prohibits processing likely detrimental to a child’s wellbeing, and prohibits tracking/behavioural monitoring of children and targeted advertising to children. ¹⁷ Your product must be built so it can operate cleanly under these constraints (especially because you are literally building a “tracking system” for minors—language alone will raise scrutiny).

Product vision and scope for the SRS

This SRS assumes you are building a **Student Mentoring and Tracking System (SMTS)** that functions as a *student development record* (mentoring + counselor notes + achievements + teacher remarks), not a full school ERP. It should still support integration/export where schools already use ERP modules for attendance/marks.

Primary objective. Provide a structured, longitudinal, role-based record of each student’s development and support interventions, aligned with India’s direction toward holistic assessment and multi-source feedback (teacher + student + parent where applicable). ¹⁸

In scope (core). Mentoring assignments, interaction logging, teacher observations/remarks, counselor session logging, student achievement and scholarship submissions, role-based access for principal/class teacher/subject teachers, and student visibility into appropriate remarks and their own records.

In scope (needed for adoption). Imports, audit trail, configurable templates, multilingual UI, offline-first support for low-connectivity environments, and export/report formats schools can share with parents and administrators. ¹⁹

Out of scope (for this SRS baseline). Fee management, transport, payroll, timetable engine, examination scheduling, and LMS/content distribution. (You can add later, but these explode scope and slow down MVP.)

Non-negotiable constraint. DPDP-compliant consent, safety, confidentiality, least-privilege access controls, and complete audit logs, because the system processes children’s personal data and counseling records.

¹⁷

Users, roles, and permissions model

Indian schools commonly use role groupings like principal, vice principal, class teacher, subject teacher, counselor/wellness teacher, and school office/admin staff. Your SRS must define permissions in a way that is configurable per school but safe by default.

Below is a **reference RBAC model** (your product must allow schools to enable/disable roles, but the default policy must be conservative around counseling notes).

Role definitions

- **System Owner (Org Admin):** Manages tenancy, contracts, data retention policies, master configurations.
- **School Admin (Principal/Authorized Admin):** School-wide configuration, user provisioning, approval workflows, reports.
- **Class Teacher:** Full access to students in their class/section for mentoring + remarks; can assign mentoring within delegated limits.
- **Mentor Teacher:** Access only to assigned mentees for mentoring logs/observations (not automatically all school students).
- **Subject Teacher:** Access to students they teach (mapped via class-section-subject allocation) to enter subject-linked remarks.
- **Counselor / Wellness Teacher:** Access to counseling cases/sessions; can publish “shareable summaries” to others based on policy.
- **Student:** Can view permitted remarks, submit achievements/scholarships, add reflections, request correction workflows.
- **Parent/Guardian (optional but recommended for India):** View-only access to published summaries and progress reports, consistent with NEP’s home-school linkage direction (and DPDP’s framing of parents/guardians in child data governance). ²⁰

Permission rules that must be explicit in the SRS

- **School-wide access is not the same as counseling access.** Counseling session notes are high-risk and should be separated from general “teacher remarks.” CBSE and NCERT emphasize counseling as a structured service; your system must treat it differently from normal classroom notes. ²¹
- **Need-to-know defaults.** A subject teacher should not automatically see counselor notes; a counselor should not automatically see all subject remarks unless assigned or consented.
- **Publishable vs confidential counseling records.** Counselors need a mechanism to create:
- **Confidential notes** (restricted),
- **Shareable summaries** (visible to defined staff + optionally parent/student),
- **Safeguarding escalation flags** (visible only to authorized safeguarding officers/principal, with audit trail).

Minimal permissions matrix (baseline)

- Principal/Admin: view all students; edit core student profile fields; cannot see confidential counselor notes unless policy grants and is logged.
- Class Teacher: view + write mentoring and general remarks for their class; view counseling *summaries* if published.

- Mentor Teacher: view + write mentoring logs for assigned mentees; view counseling summaries if published.
- Counselor: view + write counseling sessions for assigned students/cases; publish summaries; cannot edit academic records.
- Student: view published remarks; submit achievements/scholarships; cannot delete staff remarks; can request corrections.

(Where you land here is product-defining—if you get this wrong, schools will either refuse adoption or you'll create serious harm risk.)

Functional requirements for the Student Mentoring and Tracking System

This section is written in SRS “shall” format. Requirement IDs are illustrative; you can renumber later.

Identity, school setup, and user management

FR-1 School onboarding. SMTS shall support creating a School entity with identifiers including **UDISE code** (mandatory field, even if unknown at first) and board type tags. ²²

FR-2 User provisioning. SMTS shall support user creation via (a) manual entry, (b) bulk import, and (c) role-based invitation flows with verification.

FR-3 Role-based access control. SMTS shall enforce RBAC with least privilege and allow school-level configuration of role capabilities, subject to system-level safety constraints for child/counseling privacy. ¹⁷

FR-4 Auditability. SMTS shall log all access to student records, including read events for counseling records, with timestamp, user, purpose tag, and IP/device info.

Student profile and longitudinal record

FR-5 Student profile creation. Authorized roles (principal/admin/class teacher per policy) shall be able to create and update a student profile including demographic and academic placement metadata (class, section, roll/admission numbers).

FR-6 Record timeline. SMTS shall maintain a chronological timeline of all record items (mentoring interactions, counselor sessions, teacher remarks, achievements, scholarship entries), with filtering by category and academic year/term.

FR-7 Versioning. SMTS shall store immutable versions of staff-entered notes; edits shall create a new version and retain the prior version with reason-for-change.

Mentor-mentee allocation and mentoring workflow

FR-8 Mentor assignment. SMTS shall allow assigning one or more mentors to a student with effective dates and context tags (e.g., “Academic mentoring,” “Wellbeing mentoring,” “Career mentoring”).

FR-9 Capacity controls. SMTS shall support max mentees per mentor and policy controls to reflect real staffing constraints.

FR-10 Interaction logging. Mentors shall be able to record interactions with structured fields: interaction date/time, type (planned/ad-hoc), mode (in-person/phone/online), topics, observations, next steps, and follow-up date.

FR-11 Observation templates. SMTS shall provide configurable observation templates aligned with holistic assessment practices (observations over time, anecdotal records, artifacts, checklists), because NCERT's holistic assessment guidance emphasizes systematic observation and evidence capture. ²³

FR-12 Student visibility control. Each mentoring record shall have a visibility flag: (a) visible to student, (b) visible to parent/guardian, (c) staff-only, with defaults configurable.

Subject teacher allocation and remarks

FR-13 Subject mapping. SMTS shall maintain a mapping of Teacher ↔ Class/Section ↔ Subject such that a subject teacher can see only the students they teach for remark entry.

FR-14 Subject remarks. Subject teachers shall be able to write remarks linked to subject and term, with optional tagging to competencies/skills to support holistic reporting patterns. ¹⁸

FR-15 Evidence attachment. Teachers shall be able to attach artifacts (documents/images) with size limits and malware scanning.

Counselor module

FR-16 Counselor access. If a school has counselor roles (or assigns a teacher as wellness/counseling role), SMTS shall allow counselors to record sessions with structured fields: session type, presenting concerns, interventions, risk level, follow-up plan, referrals, and consent notes. ²⁴

FR-17 Confidentiality tiers. Counselor notes shall support at least three tiers: 1) Confidential session notes (counselor-only + system owner)

2) Staff shareable summary (principal/class teacher/mentor as permitted)

3) Student/parent shareable summary (published content only)

FR-18 Hub-and-spoke support. SMTS should support “cluster counselor” workflows (counselor servicing multiple schools), including monthly reporting rollups, because CBSE's hub-and-spoke model expects collaboration and reporting cycles across schools. ²⁵

FR-19 Safeguarding escalation. SMTS shall provide a “safeguarding escalation” mechanism with restricted visibility, mandatory reason entry, and immutable audit logs.

Student achievement and scholarship tracking

FR-20 Student submissions. Students shall be able to submit achievements/certificates with metadata (category, level, date, organizer, evidence attachment).

FR-21 Verification workflow. Staff shall be able to verify/approve/reject student submissions, with comments, since schools typically need authenticity control for official reporting.

FR-22 Scholarship records. Students shall be able to record scholarships applied/received, status, amount, scheme name, and supporting documents.

FR-23 NSP reference support. SMTS should include a configurable “Scholarship portal reference” field and should support capturing national portal identifiers (such as OTR on the National Scholarship Portal ²⁶) only if the school policy allows it, because NSP introduces OTR usage for scholarship applications and the system may need to store it as part of scholarship tracking. ²⁷

Principal/class teacher “enter details for all accessible students”

FR-24 Delegated data entry. SMTS shall allow principal/admin roles to enter/edit student details for all students in their scope.

FR-25 Delegation. SMTS shall support delegating specific fields to class teachers/office staff (e.g., address updates, parent contact updates) with approval requirements for sensitive changes.

Reporting and “complete tracking” outputs

FR-26 Student 360 profile. SMTS shall generate a consolidated student profile view summarizing: mentoring notes, counselor summaries (published), teacher remarks, achievements, and scholarship status across years.

FR-27 Holistic report export. SMTS shall support exports that align with NEP’s “holistic progress card” direction (multi-dimensional reporting and home-school linkage) without requiring a single rigid format, since NEP expects redesign by states/UTs and contextual implementation. ¹⁸

FR-28 Early warning indicators (configurable). SMTS should support non-invasive “attention needed” flags based on school-entered signals (e.g., repeated negative remarks, missed follow-ups), but must avoid prohibited child tracking/behavioral monitoring patterns for non-educational purposes and must remain policy-driven and transparent to authorized users. ¹⁷

Data model and workflows

Core entities (logical model)

At minimum, the data model needs these objects:

- **School** (UDISE code, board tags, location, academic calendar) ²²
- **User** (role assignments, school mapping, authentication factors)
- **Student** (current class/section, identity fields, guardians, consent state)
- **Enrollment** (academic year, class/section history)
- **SubjectAssignment** (teacher ↔ subject ↔ class/section ↔ year/term)
- **MentorAssignment** (mentor ↔ student, dates, purpose tags)
- **InteractionRecord** (mentoring session note + structured fields + visibility flags)
- **CounselingCase / CounselingSession** (structured + confidentiality tier)
- **Remark** (subject-linked or general; rubric tags; visibility policy)
- **Achievement** (submitted by student; verification status)
- **ScholarshipRecord** (applied/received; evidence; optional portal refs) ²⁷
- **Attachment** (file metadata, antivirus scan status, access control list)
- **ConsentArtifact** (what consent was given, by whom, when, scope)
- **AuditLog** (read/write events, escalations, exports)

Workflows that must be specified clearly

Onboarding workflow. 1) Create school + academic calendar.

2) Configure roles and privacy policies (especially counseling visibility).

- 3) Bulk import teachers, students, classes/sections, subject allocations.
- 4) Assign mentors and counselors.

Mentoring cycle workflow (repeatable). 1) Mentor assignment created.

- 2) Interaction recorded (structured template).
- 3) Follow-up tasks created and tracked.
- 4) Summary view updated automatically.

This matches the structured observation and evidence approach emphasized in holistic tracking guidance (observation over time, recording evidence, reflecting and planning). ²³

Counseling workflow. 1) Case created (referral source recorded: self, teacher, parent, admin).

- 2) Consent recorded where required. ¹⁷
- 3) Session logged (confidential by default).
- 4) Optional shareable summary published to defined staff/student/parent.
- 5) Escalation triggered if required (break-glass access rules + audit).

This aligns with the reality that counseling in schools is a structured service with defined roles and responsibilities, not informal “remarks.” ²⁸

Student achievement/scholarship workflow. 1) Student submits entry with evidence.

- 2) Staff verifies; status changes logged.
- 3) Entry appears in student profile + reports (if verified).

Corrections workflow. Students (and optionally parents/guardians) must be able to request correction of inaccurate personal data, and the system must log disposition. This is directly aligned with data principal rights/obligations in DPDP-style governance expectations (and will usually be demanded contractually). ¹⁷

Non-functional requirements and compliance expectations in India

Privacy, consent, and child safety

NFR-1 DPDP compliance baseline. SMTS shall support verifiable parental consent before processing personal data of users classified as children (under 18) and shall support consent logging and withdrawal handling. ²⁹

NFR-2 No advertising / no child tracking. SMTS shall not perform targeted advertising and shall not implement cross-context tracking or behavioral monitoring mechanisms prohibited for children. ¹⁷

NFR-3 Data minimization. SMTS shall allow schools to disable collection of optional identifiers and fields. (Your safest default is minimal identity data plus school-required fields.)

Confidentiality and access security

NFR-4 Encryption. SMTS shall encrypt data in transit and at rest; counseling notes should use additional key segregation (separate encryption context) due to higher sensitivity. ³⁰

NFR-5 Strong audit trails. SMTS shall provide immutable audit logs for viewing/exporting counseling records.

Availability and offline-first usability

Your product must work in mixed-connectivity environments. UDISE+ materials explicitly reflect offline-to-online upload patterns for remote areas and block-level support; you should design similarly for resilience.

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NFR-6 Offline capture. SMTS mobile apps shall support offline draft capture for mentoring notes and queued sync with conflict resolution.

Performance and scale

NFR-7 Typical scale targets. SMTS should support schools from <200 to >5000 students per tenant, and multi-school groups. UDISE+ scale references show India's overall system is huge; even if you serve a fraction, your architecture should not assume "small data." 32

Recordkeeping alignment

CBSE affiliation expectations demonstrate that schools must preserve and maintain multiple categories of records, including exam-related records and attendance registers, and submit e-returns. 10 Even if your product is not an exam system, your design should make it easy to export records and prove integrity (timestamps, versioning, audit logs).

Governance and community structures

Government and aided schools also have mandated governance bodies like School Management Committees in many contexts (composition and meeting/minutes expectations are documented in RTE-related rules). If you later add parent/community modules, they must map to these structures rather than inventing something incompatible. 33

Acceptance criteria and traceability outline

To make this SRS implementable, you need acceptance criteria that prove the system works under real school workflows and under privacy constraints.

Minimum acceptance criteria for MVP

- A principal can onboard a school, bulk import students/teachers, and assign mentors to students, and the mentor can record interactions using a structured template with evidence attachments. (Template approach aligns with holistic observation/evidence guidance.) 23
- A subject teacher can see only students they teach and can add subject-linked remarks; those remarks appear in the student 360 view.
- A counselor can create sessions that are confidential by default and can publish a separate shareable summary visible to the class teacher and student (only if permitted). 24
- Audit logs show who accessed counseling records; exports are logged and watermarkable.
- Student can submit an achievement and scholarship detail with evidence; staff can verify; verified items appear in reports; unverified items remain clearly marked. 27
- Consent artifacts exist for child users and can be demonstrated during a compliance review, including withdrawal handling. 17

Traceability structure

Maintain a traceability mapping from: - Policy drivers (NEP holistic progress expectations; counseling ecosystem requirements; DPDP child protections) → functional modules → requirements → test cases. 34

If you don't build traceability early, you will lose control of scope and you won't be able to defend your design choices when a school (or a regulator/partner) asks why certain access controls exist.

1 2 3 4 14 18 20 34 **education.gov.in**

https://www.education.gov.in/sites/upload_files/mhrd/files/NEP_Final_English_0.pdf

5 8 9 22 **UDISE + Reports**

<https://udiseplus.gov.in/udisereport/>

6 32 **UDISE+**

<https://udiseplus.gov.in/ud/homePage/index.jsp>

7 19 31 **dsel.education.gov.in**

https://dsel.education.gov.in/sites/default/files/update/UDISE_Booklet.pdf

10 **CBSE - Central Board of Secondary Education**

<https://www.cbse.gov.in/cbsenew/aff-bye-laws.html>

11 **Boards Expectations from Teachers | Central Board of Secondary Education**

<https://www.cbse.gov.in/cbsenew/cbse-exp-teacher.html>

12 24 **saras.cbse.gov.in**

https://saras.cbse.gov.in/saras/Circulars/Circular02_2026639044402638061650.pdf

13 21 25 26 **cbse.gov.in**

https://www.cbse.gov.in/cbsenew/documents/CBSE_Counseling_Hub_Spoke_School_Model_28082025%20.pdf

15 28 **ncert.nic.in**

https://ncert.nic.in/depfe/pdf/Guidelines_for_Guidance_and_Counseling.pdf

16 17 29 30 **meity.gov.in**

<https://www.meity.gov.in/static/uploads/2024/06/2bf1f0e9f04e6fb4f8fef35e82c42aa5.pdf>

23 **cbseacademic.nic.in**

https://cbseacademic.nic.in/web_material/Manuals/HPC_TeacherGuide.pdf

27 **NSP : National Scholarship Portal**

<https://scholarships.gov.in/>

33 **righttoeducation.in**

<https://righttoeducation.in/sites/default/files/>

[Right%20of%20Children%20to%20Free%20and%20Compulsory%20Education%20Act%202009%20%28English%29.pdf](#)